

NATIONAL UNIVERSITY OF TECHNOLOGY
(Department of Computer Science)

CS280– Artificial Intelligence

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Assignment 1

Total Marks: 40

Question 1 [Marks: 10, CLO: 1, PLO: 2]

Artificial Intelligence (AI) is transforming disaster management by predicting hazards, analysing real-time data, and assisting rescue teams in making faster and more accurate decisions. In this case study, you will design the conceptual framework of an AI-Powered Disaster Response System. Acting as AI system designers, outline the system using the PEAS model (Performance, Environment, Actuators, Sensors), analyse its interaction with the environment, and classify its agent type and behaviour.

1. Agent Perception and Disaster Tracking

a. Percept: What data does the system collect? The system collects real-time and historical data from multiple sources:

- Satellite imagery for large-scale environmental monitoring
- IoT sensors for seismic activity, temperature, humidity, and flood levels
- Social media feeds for crowd-sourced updates and emergency signals
- Emergency service databases for incident reports and resource status
- GPS and mobile data for population movement and density tracking

b. States: What disaster states does the system recognize? The system identifies and transitions between the following states:

- Normal: Routine monitoring, no immediate threat
- Alert: Potential hazard detected, early warnings issued
- Active: Disaster is occurring, emergency protocols activated
- Recovery: Post-disaster phase, damage assessment and aid coordination

2. PEAS Framework for AIDRS

Component	Description
Performance Measure	Accuracy of hazard prediction, speed of response, reduction in casualties, efficiency of resource allocation
Environment	Disaster-prone regions, urban and rural infrastructure, weather systems, population mobility patterns
Actuators	Alert systems (sirens, SMS, notifications), drones and robots, emergency dispatch, resource allocation systems
Sensors	Satellite feeds, IoT devices, surveillance cameras, social media APIs, GPS data

3. Classification of AIDRS Environment

Aspect	Classification	Justification
Static or Dynamic	Dynamic	Disaster conditions and human responses change rapidly
Deterministic or Stochastic	Stochastic	Outcomes are uncertain due to natural forces and human behavior
Episodic or Sequential	Sequential	Each decision affects future states and must consider historical context
Fully or Partially Observable	Partially Observable	Not all data is visible or accessible (e.g., hidden casualties)
Discrete or Continuous	Continuous	Environmental data changes fluidly over time

Single-Agent or Multi-Agent	Multi-Agent	Involves coordination with multiple stakeholders and systems
Known or Unknown	Unknown	May face novel disasters or conditions not previously encountered

4. Type of Intelligent Agent

AIDRS is best classified as:

- **Utility-Based Agent:** It evaluates multiple actions and selects the one that maximizes overall benefit (e.g., saving lives, minimizing damage).
- **Learning Agent:** It improves over time by learning from past disasters and refining its prediction and response strategies.

5. AI System Capabilities

- Thinks Humanly Mimics human reasoning by analyzing patterns, interpreting social signals, and learning from experience.
- Acts Humanly Communicates with people via natural language alerts, interprets human-generated data, and supports decision-making.
- Thinks Rationally Uses logic and probabilistic models to evaluate risks, prioritize actions, and optimize outcomes.
- Acts Rationally Executes decisions that maximize utility under uncertainty, such as selecting optimal evacuation routes or allocating resources efficiently.

Question 2 [Marks: 5, CLO: 1, PLO: 2]

Let p stand for the proposition “I bought a lottery ticket”, and q for “I won the jackpot”. Express the following as natural English sentences:

1. $p \vee q$

Translation: I bought a lottery ticket or I won the jackpot.

This means at least one of the two statements is true.

2. $p \wedge q$

Translation: I bought a lottery ticket and I won the jackpot.

Both events occurred.

3. $p \Rightarrow q$

Translation: If I bought a lottery ticket, then I won the jackpot.

This implies that buying the ticket led to winning.

4. $\neg p \Rightarrow \neg q$

Translation: If I did not buy a lottery ticket, then I did not win the jackpot. This suggests that not buying a ticket guarantees not winning.

5. $\neg p \vee (p \wedge q)$

Translation: Either I did not buy a lottery ticket, or I bought a ticket and won the jackpot.

This excludes the possibility of buying a ticket and not winning.

Question 3 [Marks: 5, CLO: 1, PLO: 2]

Formalise the following in terms of atomic propositions r , b , and w , first making clear how they correspond to the English text.

ANS:

- r : Rabbits have been seen in the area
- b : Berries are ripe along the path
- w : Walking on the path is safe

1. Berries are ripe along the path, but rabbits have not been seen in the area.

Formalization: $b \wedge \neg r$

2. Rabbits have not been seen in the area, and walking on the path is safe, but berries are ripe along the path.

Formalization: $\neg r \wedge w \wedge b$

3. If berries are ripe along the path, then walking is safe if and only if rabbits have not been seen in the area.

Formalization: $b \Rightarrow (w \Leftrightarrow \neg r)$

4. It is not safe to walk along the path, but rabbits have not been seen in the area and the berries along the path are ripe.

Formalization: $\neg w \wedge \neg r \wedge b$

5. Walking is not safe on the path whenever rabbits have been seen in the area and berries are ripe along the path.

Formalization: $(r \wedge b) \Rightarrow \neg w$

Question 4 [Marks: 5, CLO: 1, PLO: 2]

For each of the following propositions, construct a truth table and state whether the proposition is valid or satisfiable.

1. $p \wedge \neg p$

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

- **Valid?** No
- **Satisfiable?** No
- **Explanation:** This is a contradiction. It's never true.

2. $p \vee \neg p$

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

- **Valid?** Yes
- **Satisfiable?** Yes
- **Explanation:** This is a tautology. It's always true.

3. $(p \vee \neg q) \Rightarrow q$

p	q	$\neg q$	$p \vee \neg q$	$(p \vee \neg q) \Rightarrow q$
T	T	F	T	T
T	F	T	T	F

F	T	F	F	T
F	F	T	T	F

- **Valid?** No
- **Satisfiable?** Yes
- **Explanation:** False in some cases, but true in others.

4. $(p \vee q) \Rightarrow (p \wedge q)$

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \Rightarrow (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

- **Valid?** No
- **Satisfiable?** Yes
- **Explanation:** Not always true, but true in some cases.

5. $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$

p	q	$\neg p$	$\neg q$	$p \Rightarrow q$	$\neg q \Rightarrow \neg p$	$\Leftrightarrow$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

- **Valid?** Yes
- **Satisfiable?** Yes
- **Explanation:** This is logically equivalent to the contrapositive. Always true.

6. $(p \Rightarrow q) \Rightarrow (q \Rightarrow p)$

p	q	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \Rightarrow (q \Rightarrow p)$
T	T	T	T	T

T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

- **Valid?** No
- **Satisfiable?** Yes
- **Explanation:** Not always true, fails when p is false and q is true.

Question 5 [Marks: 2, CLO: 1, PLO: 2]

Formalize the following and, by writing truth tables for the premises and conclusion, determine whether the arguments are valid.

1. Premises:

- Either John isn't stupid and he is lazy, or he's stupid.
- John is stupid.

Conclusion: Therefore, John isn't lazy.

ANS:

Propositions

- Let **s** = "John is stupid"
- Let **l** = "John is lazy"

Premises

- $(\neg s \wedge l) \vee s$
- s

Conclusion

$\neg l$

Truth Table

s	l	$\neg s$	$\neg l$	$(\neg s \wedge l) \vee s$	Conclusion ($\neg l$)
T	T	F	F	T	F

T	F	F	T	T	T
F	T	T	F	T	F
F	F	T	T	F	T

Analysis

- In row 1 and 2, both premises are true.
- In row 1, conclusion is false → **invalid**
- In row 2, conclusion is true → **satisfiable**

Conclusion: The argument is **not valid** (not true in all cases), but it is **satisfiable** (true in some cases).

2. Premises:

a. Either the butler is lying or the cook is innocent.

b. Therefore, the butler is either lying or guilty.

Conclusion: The butler and the cook are not both innocent

ANS:**Propositions**

- Let b = "The butler is lying"
- Let c = "The cook is innocent"

Premises

1. $b \vee c$
2. $b \vee \neg c$

Conclusion

$\neg(b \wedge c)$

Truth Table

b	c	$\neg c$	$b \vee c$	$b \vee \neg c$	$b \wedge c$	$\neg(b \wedge c)$
T	T	F	T	T	T	F
T	F	T	T	T	F	T

F	T	F	T	F	F	T
F	F	T	F	T	F	T

Analysis

- In row 1, both premises are true, but conclusion is false $\rightarrow$ invalid
- In row 2, both premises and conclusion are true $\rightarrow$ satisfiable

Conclusion: The argument is not valid, but it is satisfiable.

Question 6 [Marks: 5, CLO: 1, PLO: 2]

Formalize the following statements in predicate logic, making clear what your atomic predicate symbols stand for and what the domains of any variables are.

1. Anyone who has forgiven at least one person is a saint.**Predicate Symbols:**

- Forgives(x, y): Person x has forgiven person y
- Saint(x): Person x is a saint

Formalization: $\forall x [(\exists y \text{ Forgives}(x, y)) \rightarrow \text{Saint}(x)]$

2. Nobody in the calculus class is smarter than everybody in the discrete math class.**Predicate Symbols:**

- Calc(x): Person x is in the calculus class
- Disc(y): Person y is in the discrete math class
- Smarter(x, y): Person x is smarter than person y

Formalization: $\neg \exists x [\text{Calc}(x) \wedge \forall y (\text{Disc}(y) \rightarrow \text{Smarter}(x, y))]$

3. Anyone who has bought a Rolls Royce with cash must have a rich uncle.**Predicate Symbols:**

- BoughtWithCash(x, r): Person x bought item r with cash
- RollsRoyce(r): Item r is a Rolls Royce
- Uncle(u, x): Person u is the uncle of person x
- Rich(u): Person u is rich

Domain:

- x, u : people
- r : items (cars)

Formalization: $\forall x [(\exists r (\text{RollsRoyce}(r) \wedge \text{BoughtWithCash}(x, r))) \rightarrow \exists u (\text{Uncle}(u, x) \wedge \text{Rich}(u))]$

4. If anyone in the college has the measles, then everyone who has a friend in the college will have to be quarantined.

Predicate Symbols:

- $\text{InCollege}(x)$: Person x is in the college
- $\text{HasMeasles}(x)$: Person x has the measles
- $\text{Friend}(y, z)$: Person y is a friend of person z
- $\text{Quarantined}(y)$: Person y must be quarantined

Formalization: $(\exists x (\text{InCollege}(x) \wedge \text{HasMeasles}(x))) \rightarrow \forall y [(\exists z (\text{InCollege}(z) \wedge \text{Friend}(y, z))) \rightarrow \text{Quarantined}(y)]$

5. If Jones can do it, anyone can.

Predicate Symbols:

- $\text{CanDo}(x)$: Person x can do it
- Let j be a constant representing Jones

Formalization: $\text{CanDo}(j) \rightarrow \forall x \text{CanDo}(x)$

Question 7 [Marks: 3, CLO: 1, PLO: 2]

Translate the following into idiomatic English.

1. $\forall x. (H(x) \wedge \forall y. \neg M(x, y)) \Rightarrow U(x)$ where $H(x)$ means x is a man, $M(x, y)$ means x is married to y , $U(x)$ means x is unhappy, and x and y range over people.

ANS:

Translation: Every man who is not married to anyone is unhappy.

Explanation: This statement says that if someone is a man and is not married to any person, then he is unhappy.

2. $\exists z. P(z, x) \wedge S(z, y) \wedge W(y)$ where $P(z, x)$ means z is a parent of x , $S(z, y)$ means z and y are siblings, $W(y)$ means y is a woman, and x , y , and z range over people.

ANS:

Translation: There exists someone who is a parent of x , is a sibling of a woman y .

Explanation: This means that there is a person z who is both a parent of x and a sibling of some woman y .

Question 8 [Marks: 5, CLO: 1, PLO: 2]

State whether the following are true or false, where x , y and z range over the integers.

1. $\forall x \exists y (2x - y = 0)$

Translation: For every integer x , there exists an integer y such that $y = 2x$.

- This is **true**, because for any integer x , we can always find $y = 2x$, which is also an integer.

Answer: **True**

2. $\exists y \forall x (2x - y = 0)$

Translation: There exists an integer y such that for every integer x , $2x = y$.

- This is **false**, because $2x$ varies with x . No single y can satisfy $2x = y$ for all x .

Answer: **False**

3. $\forall x \exists y (x - 2y = 0)$

Translation: For every integer x , there exists an integer y such that $y = x/2$.

- This is **false**, because $x/2$ is not always an integer (e.g., $x = 1$ gives $y = 0.5$).

Answer: **False**

4. $\forall x (x < 10 \Rightarrow \forall y (y < x \Rightarrow y < 9))$

Translation: For every integer x less than 10, if y is less than x , then y is less than 9.

- Consider $x = 9$. Then $y < 9$ implies $y < 9$, which is fine.
- But if $x = 8$, then $y < 8 \Rightarrow y < 9$ is always true.
- The implication holds for all $x < 10$.

Answer: **True**

5. $\exists y \exists z (y + z = 100)$

Translation: There exist integers y and z such that their sum is 100.

- This is **true**, for example $y = 50$, $z = 50$.

Answer: **True**

